

Mathematics Stage 4

Expanding Brackets

Book 4

Version: 260930

Feedback:
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Contents

Syllabus Content	2
Expanding Brackets 1	3
Expanding Brackets 2	10
Challenge Exercise	16

Syllabus Content

MA4-ALG-C-01 generalises number properties to operate with algebraic expressions including expansion and factorisation

Extend and apply the distributive law to the expansion of algebraic expressions

- Explain the role and meaning of grouping symbols in algebraic expressions
- Apply the distributive law to expand and simplify algebraic expressions by removing grouping symbols

Expanding Brackets 1

Review

☐ Simplify algebraic multiplication

a $3 \times 2a$

b $4 \times 3x$

c $5(2b)$

d $2(4y)$

☐ Multiply integers

a $3 \times (-4)$

b $5 \times (-2)$

c 3×4

d -2×5

e $-3(-4)$

f $-5(-2)$

g $-5(4)$

h $-2(-3)$

☐ Simplify algebraic multiplication

a $3(-2a)$

b $4(-3x)$

c $-2(3b)$

d $-3(4x)$

e $-2(-3a)$

f $-4(-2x)$

g $6(-2y)$

h $-12(-2xy)$

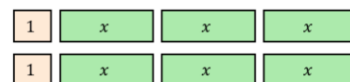
☐ Multiply grouped terms

Using the diagrams, multiply these grouped terms.

a $3 \times (4 + x)$



b $2(1 + 3x)$



Expanding Brackets

We **group** terms using **brackets**.

Expanding brackets means simplifying an expression by **writing without brackets**.

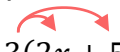
To expand brackets, we multiply each term inside the brackets by the term outside.

This is called the **distributive law**.

$$a(b + c) = ab + ac$$


Example Expand brackets using the distributive law

Expand:

$$\begin{aligned} & 3(2x + 5) \\ = & 3(2x) + 3(5) \\ = & 6x + 15 \end{aligned}$$


To expand a bracket, multiply e..... term inside the brackets by the term o.....

Guided Practice

Expand:

a $4(3a + 2)$



b $4(3a - 2)$



c $5(x - 3)$

d $-2(3x + 4)$

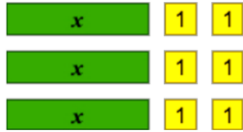
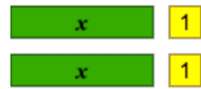
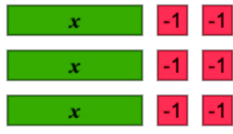
e $-3(2x - 5)$

f $2(a + 3b + 4)$

Key Ideas

1. To expand a bracket, multiply term inside the brackets by the term the bracket.
2. When there is subtraction inside the brackets, such as $4(3a - 2)$, treat the two terms inside the bracket as and

1 Complete the table.

Expression	Diagram	Expanded Expression
$3(x + 2)$		$3x + 6$
$2(x + 1)$		$2x + 2$
$2(x - 2)$		$2x - 4$
$2(x + 4)$		$2x + 8$
$3(x + 1)$		$3x + 3$
$3(x - 2)$		$3x - 6$

2 Expand the brackets.

a $3(x + 4)$

$3x + 12$

b $3(x - 4)$

$3x - 12$

c $3(2x + 4)$

$6x + 12$

d $3(2x - 4)$

$6x - 12$

e $3(2x - 5y + 4)$

$6x - 15y + 12$

3 Expand these expressions using the distributive law.

a $6(y + 8)$

$6y + 48$

b $8(s + 7)$

$8s + 56$

c $-7(x + 5)$

$-7x - 35$

d $9(x - 9)$

$9x - 81$

e $8(y - 4)$

$8y - 32$

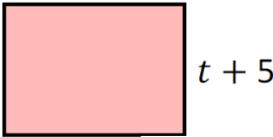
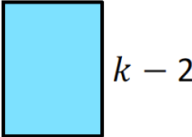
f $-6(3 - e)$

$-18 + 6e$

4 Expand these expressions using the distributive law.

a $10(6g - 7)$	b $5(7w + 10)$	c $-7(8x - 2)$
$60g - 70$	$35w + 50$	$-56x + 14$
d $7(q - 7)$	e $2(2u - 6)$	f $-6(8l + 5)$
$7q - 49$	$4u - 12$	$-48l - 30$

5 The area of a rectangle is $A = \text{length} \times \text{width}$.
Write an expression for the area of these rectangles.

a  $4(2x - 5) = \dots\dots\dots$ $8x - 20$	b  $5t + 25$	c  $7k - 14$
---	--	--

6 Expand these brackets.

a $2(2b - 3)$	b $-5(2x + 3)$	c $2(3x + 4)$	d $-3(2a - 5)$
$4b - 6$	$-10x - 15$	$6x + 8$	$-6a + 15$
e $3(4x + 2)$	f $-2(3x + 4)$	g $4(3a - 2)$	h $-5(2x - 3)$
$12x + 6$	$-6x - 8$	$12a - 8$	$-10x + 15$
i $4(3b + 5)$	j $-2(3x - 4)$	k $5(2x - 4)$	l $-3(2a + 5)$
$12b + 20$	$-6x + 8$	$10x - 20$	$-6a - 15$
m $3(2x - 5)$	n $-4(3b + 2)$	o $5(2a + 3)$	p $-4(3b - 2)$
$6x - 15$	$-12b - 8$	$10a + 15$	$-12b + 8$

7

a The side length of a square is $x + 2$. Write an expression for the perimeter.	b The side length of a regular hexagon is $2x + 5$. Write an expression for the perimeter.
$4(x + 2)$ or $4x + 8$	$6(2x + 5)$ or $12x + 30$

8 Write an expression for each of the following and then expand it.

a A number, x , has 3 added to it and the result is multiplied by 5.

$$5x + 15$$

b A number, b , has 6 added to it and the result is doubled.

$$2b + 12$$

c A number, z , has 4 subtracted from it and the result is multiplied by 3.

$$3z - 12$$

d A number, y , is subtracted from 10 and the result is multiplied by 7.

$$70 - 7y$$

9 The number of boys in a classroom is b and the number of girls is g . Each boy has 5 pencils and each girl has 3 pencils.

a Write an expression for the total number of pencils in the class.

$$5b + 3g$$

b If the pencils cost \$2 each, write and expand an expression for the total cost of all the pencils in the room.

$$10b + 6g$$

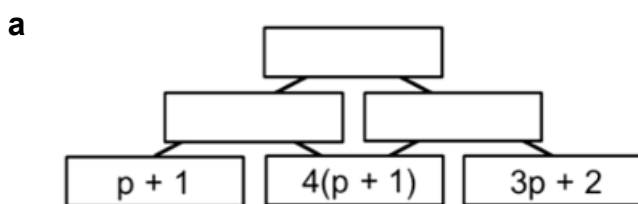
c Each boy and girl also has one pencil case, costing \$4 each. Write a simplified and expanded expression for the total cost of all pencils and cases.

$$14b + 10g$$

d If there are 10 boys and eight girls in the room, what is the total cost for all the pencils and cases in the room?

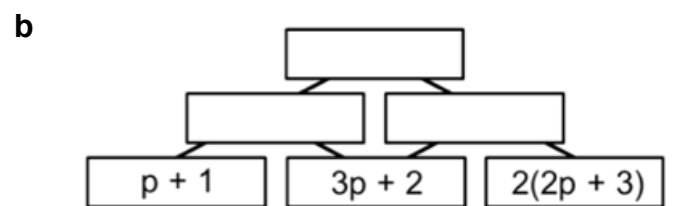
$$220$$

10 Complete each pyramid. Each expression is the sum of the two expressions below it.



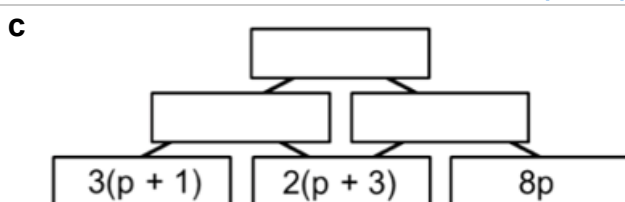
$$12p + 11$$

$$5p + 5, 7p + 6$$



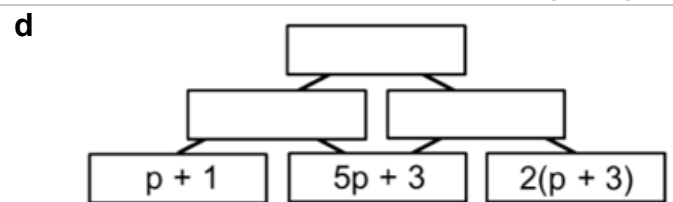
$$11p + 11$$

$$4p + 3, 7p + 8$$



$$15p + 15$$

$$5p + 9, 10p + 6$$



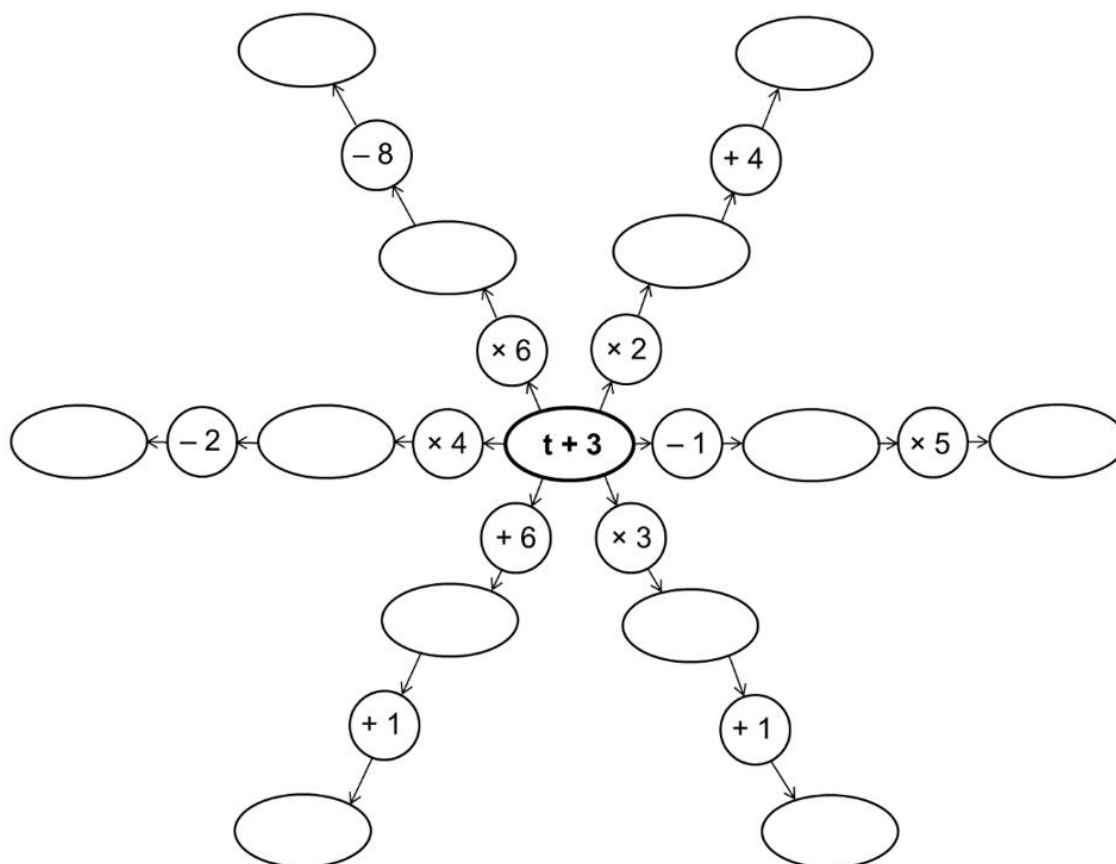
$$13p + 13$$

$$6p + 4, 7p + 9$$

11 Complete these sequences of operations.

- a** $k + 2 \rightarrow +4 \rightarrow \square \rightarrow \times 3 \rightarrow \square$ $k + 6, 3k + 18$
- b** $k + 2 \rightarrow \times 3 \rightarrow \square \rightarrow +4 \rightarrow \square$ $3k + 6, 3k + 10$
- c** $2m + 1 \rightarrow \times 2 \rightarrow \square \rightarrow +3 \rightarrow \square$ $4m + 2, 4m + 5$
- d** $2m + 1 \rightarrow +3 \rightarrow \square \rightarrow \times 2 \rightarrow \square$ $2m + 4m, 4m + 8$
- e** $6y + 9 \rightarrow \div 3 \rightarrow \square \rightarrow +12 \rightarrow \square$ $2y + 3, 2y + 15$
- f** $6y + 9 \rightarrow +12 \rightarrow \square \rightarrow \div 3 \rightarrow \square$ $6y + 21, 2y + 7$

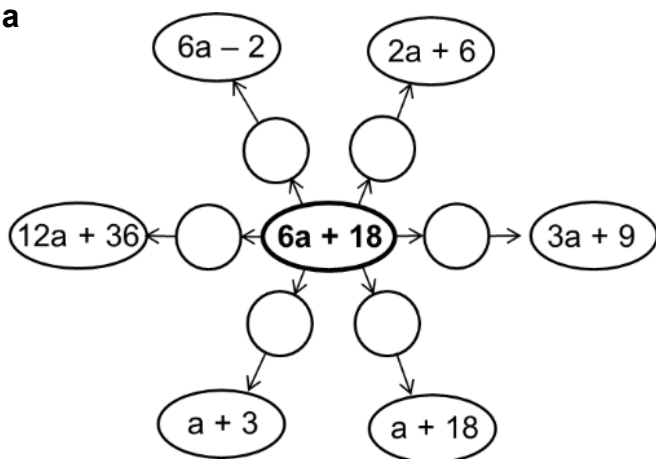
12 Start in the middle and work outwards. Fill in the missing expressions.



(outer clockwise from top left): $6t + 10, 2t + 10, 5t + 10, 3t + 10, t + 10, 4t + 10$

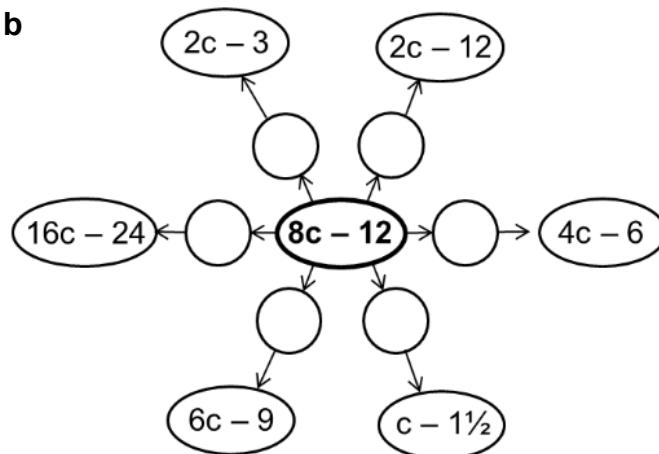
13 Start in the middle and work outwards. What are the missing operations?

a



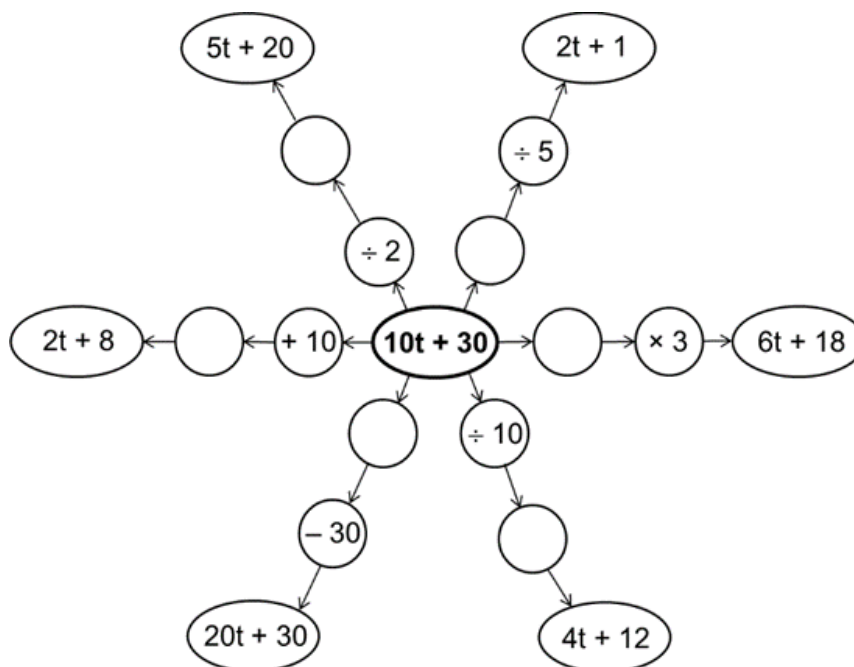
(clockwise from top left): $\div 2$, $-3b$, $\times 2$, -10 , $\div 4$, $\times 5$

b



(clockwise from top left): $\div 4$, $-6c$, $\div 2$, $\div 8$, $\times \frac{3}{4}$, $\times 2$

14 Start in the middle and work outwards. What are the missing operations?



(clockwise from top left): $+5$, -25 , $\div 5$, $\times 4$, $\times 2$, $\div 5$

15 Expand these expressions involving fractions.

a

$$\frac{1}{3}(6x + 15)$$

$$2x + 5$$

b

$$\frac{1}{4}(12x + 8y)$$

$$3x + 2y$$

c

$$\frac{3}{4}(16 - 12y)$$

$$12 - 8y$$

d

$$\frac{4x + 10y}{2}$$

$$2x + 5y$$

Expanding Brackets 2

Review

☐ Simplify algebraic multiplication

a $x \times y$

b $a \times b$

c $p \times q$

d $x \times 2y$

e $4y \times x$

f $a \times 3b$

g $2p \times 3q$

h $x \times x$

i $3x \times x$

j $3ab \times a$

k $2xy \times y$

l $5pq \times 3p$

☐ Expand brackets

a $3(4a - b)$

b $3(x - 2)$

c $12(2x + 3)$

d $-2(x - y)$

Expanding Brackets

To expand brackets, we multiply each term inside the brackets by the term outside.

$$a(b + c) = ab + ac$$


Example Expand brackets using the distributive law

Expand $x(2y + 3)$

$$= x(2y) + x(3)$$

$$= 2xy + 3x$$

To expand a bracket, multiply e..... term inside the brackets by the term o.....

Expand $x(x + 4)$

$$= x(x) + x(4)$$

$$= x^2 + 4x$$

When something is multiplied by itself, we write $x \times x = \dots\dots$

Guided Practice

Expand:

a $a(3b - 2)$

b $x(x - 5)$

c $2x(x + 3)$

d $3x(2x - 4)$

e $3xy(2x - 5y)$

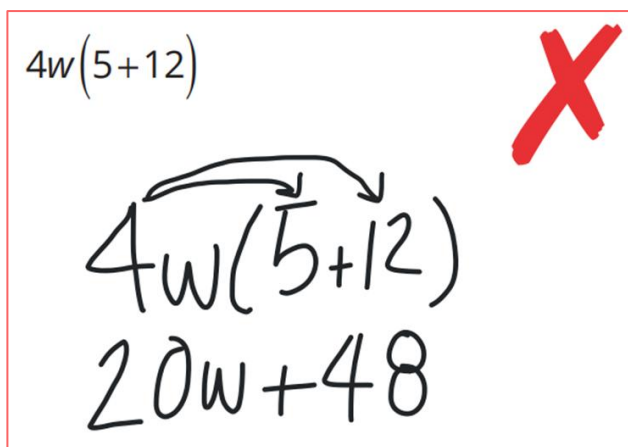
f $-3y(4 - 5y)$

Key Ideas

1. To expand a bracket, multiply the term inside the brackets by the term the bracket.
2. When we multiply two of the same variable together, we write:
 $x \times x = \dots\dots$, and $ab \times a = \dots\dots$

1 Read the example carefully and answer the questions.

Pablo's work:



$$4w(5+12)$$

$$20w + 48$$

What did Pablo forget when distributing the $4w$?

He forgot to write the with 48.

What should Pablo's final expression be?

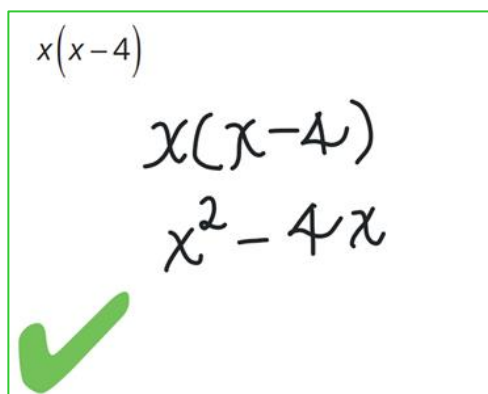
$68w$

Expand the following: $4x(3 + 11)$

$56x$

2 Read the examples carefully and answer the questions.

Michael's work:



$$x(x-4)$$

$$x^2 - 4x$$

Where did x^2 come from in Michael's answer?

..... $\times$ = x^2

Expand the following: $3x(x - 4)$

$3x^2 - 12x$

3 Expand the brackets.

a $a(a + 4)$	b $a(b + 4)$	c $a(b - 4)$	d $a(4 - b)$	e $a(4 - a)$
$a^2 + 4a$	$ab + 4a$	$ab - 4a$	$4a - ab$	$4a - a^2$
f $a(3a - 4)$	g $a(4 - 3a)$	h $3a(a - 4)$	i $3a(2a - 4)$	j $3a(4 - 2a + 5b)$
$3a^2 - 4a$	$4a - 3a^2$	$3a^2 - 12a$	$6a^2 - 12a$	$12a - 6a^2 + 15ab$

4 Expand these brackets.

a $3a(a + 2)$	b $x(4y - 3)$	c $2x(3x + 4)$	d $b(b - 2)$
$3a^2 + 6a$	$4xy - 3x$	$6x^2 + 8x$	$b^2 - 2b$
e $2xy(3x - 2y)$	f $a(3b + 2)$	g $4x(2x - 3)$	h $y(y - 4)$
$6x^2y - 4xy^2$	$3ab + 2a$	$8x^2 - 12x$	$y^2 - 4y$
i $3ab(2a + 5)$	j $2x(x - 5)$	k $a(2b - 5)$	l $x(x + 3)$
$6a^2b + 15ab$	$2x^2 - 10x$	$2ab - 5a$	$x^2 + 3x$
m $4ab(2a - 3b)$	n $2a(5a - 4)$	o $x(2y + 5)$	p $4a(a - 3)$
$8a^2b - 12ab^2$	$10a^2 - 8a$	$2xy + 5x$	$4a^2 - 12a$
q $2xy(3x + 4)$	r $a(5 + a)$	s $3a(2a + 5)$	t $2x(4 + x)$
$6x^2y + 8xy$	$5a + a^2$	$6a^2 + 15a$	$8x + 2x^2$

5 Expand these expressions.

a $x(y - 3)$	b $d(x + 9s)$	c $5a(2x + 3)$
$xy - 3x$	$dx + 9ds$	$10ax + 15a$
d $5c(2w - 4)$	e $-7y(4x + 7)$	f $5j(r + 7p)$
$10cw - 20c$	$-28xy - 49y$	$5jr + 35jp$
g $-8d(t - 3d)$	h $-7k(3x + 2)$	i $14k(y + 1)$
$-8dt + 24d^2$	$-21kx - 14k$	$14ky + 14k$

6 Fill in the empty boxes.

a $\square(x + 4) = 2x + 8$	b $\square(2x - 3) = 8x - 12$
2	4
c $\square(2x + 3) = 6x^2 + 9x$	d $4(\square + 5) = 12x + 20$
3x	3x
e $4y(\square - \square) = 4y^2 - 4y$	f $-2x(\square + \square) = -4x^2 - 6xy$
y, 1	2x, 3y

7 Expand these expressions, then simplify the result.

a $2(x + 1) + 3(x + 4)$

b $3(x + 4) + 2(x + 1)$

c $x(x - 1) + 1(x - 1)$

d $a(b + 3) + 2(b + 3)$

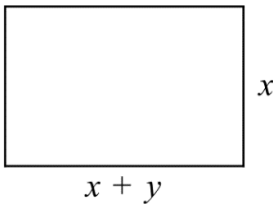
$5x + 14$

$5x + 14$

$x^2 - 1$

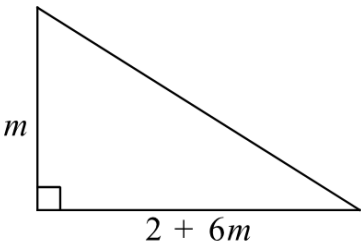
$ab + 3a + 2b + 6$

8 Write an algebraic expression and expand it to represent the area of the rectangle.



$x^2 + xy$

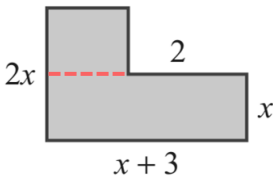
9 Write a fully simplified expression to represent the area of the triangle.



$m + 3m^2$

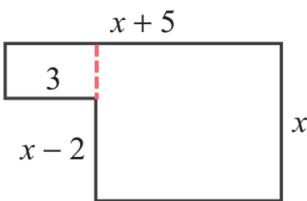
10 Find the area of these shapes in simplest expanded form.

a



$2x^2 + 4x$

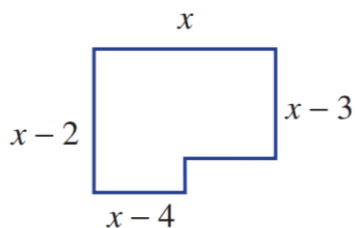
b



$6 + x^2 + 2x$

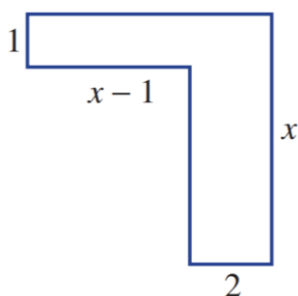
11 Find expressions, in simplest form, for the perimeter (P) and area (A) of these shapes.

a



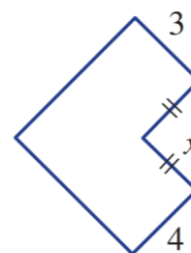
$$\begin{aligned} P: & 4x - 4 \\ A: & x^2 - 2x - 4 \end{aligned}$$

b



$$\begin{aligned} P: & 4x + 2 \\ A: & 3x - 1 \end{aligned}$$

c



$$\begin{aligned} P: & 4x + 14 \\ A: & 7x + 4 \end{aligned}$$

12 Expand the following. Be careful of expanding negatives.

a $5a(a - 3) - 2a(3a - 1)$

$$\begin{aligned} &= 5a^2 - 15a - 6a^2 + 2a \\ &= -a^2 - 13a \end{aligned}$$

b $4y(2y + 3) - 3y(y - 5)$

$$\begin{aligned} &= 8y^2 + 12y - 3y^2 + 15y \\ &= 5y^2 + 27y \end{aligned}$$

13 Expand the following:

a $\frac{x}{2}(4x + 6) + \frac{x}{3}(6x - 9)$

$$\begin{aligned} &= 2x^2 + 3x + 2x^2 - 3x \\ &= 4x^2 \end{aligned}$$

b $\frac{3a}{4}(8a - 4) - \frac{2a}{3}(6a + 3)$

$$\begin{aligned} &= 6a^2 - 3a - 4a^2 - 2a \\ &= 2a^2 - 5a \end{aligned}$$

c $\frac{2x}{3}(3x + 6) + \frac{x}{4}(8x - 12)$

$$\begin{aligned} &= 2x^2 + 4x + 2x^2 - 3x \\ &= 4x^2 + x \end{aligned}$$

d $\frac{y}{5}(10y - 15) + \frac{2y}{5}(5y + 10)$

$$\begin{aligned} &= 2y^2 - 3y + 2y^2 + 4y \\ &= 4y^2 + y \end{aligned}$$

Challenge Exercise

1

- a A car travels at 60 km/h for 3 hours. What is the distance travelled in the 3 hours?

$$60 \times 3 = 180 \text{ km}$$

- b A car travels at 60 km/h for n hours. What is the distance travelled in the n hours?

$$60n \text{ km}$$

- c A car travels at x km/h for 3 hours. What is the distance travelled in the 3 hours?

$$3x \text{ km}$$

- d A car travels at x km/h for n hours. What is the distance travelled in the n hours?

$$xn \text{ km}$$

2

- a A car travels at a constant speed for 2 hours. It travels 120 km in this time. What is the speed of the car?

$$\frac{120}{2} = 60 \text{ km/h}$$

- b A car travels at a constant speed for 3 hours. It travels n km in this time. What is the speed of the car?

$$\frac{n}{3} \text{ km/h}$$

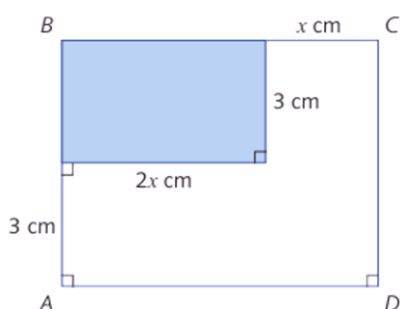
- c A car travels at a constant speed for m hours. It travels n km in this time. What is the speed of the car?

$$\frac{n}{m} \text{ km/h}$$

- 3 If a car is travelling at 50 km/h, how long does it take for the car to travel n km?

$$\frac{n}{50} \text{ h}$$

- 4 For the diagram below:



- a Find the areas of rectangle ABCD and the shaded rectangle.

$$ABCD: 3x \times 6 = 18x \text{ cm}^2$$

$$\text{Shaded: } 2x \times 3 = 6x \text{ cm}^2$$

- b Find the difference of the two areas found in part a.

$$18x - 6x = 12x \text{ cm}^2$$

5

- a A rectangle has area 56 cm^2 and width 7 cm . What is the length of the rectangle?

$$\frac{56}{7} = 8 \text{ cm}$$

- b A rectangle has area $x \text{ cm}^2$ and width 3 cm . What is the length of the rectangle?

$$\frac{x}{3} \text{ cm}$$

- c A rectangle has area $(x + 4) \text{ cm}^2$ and width 3 cm . What is the length of the rectangle?

$$\frac{x + 4}{3} \text{ cm}$$

- d A rectangle has area 3 cm^2 and width $x \text{ cm}$. What is the length of the rectangle?

$$\frac{3}{x} \text{ cm}$$

6

- a A rectangle has area $x \text{ cm}^2$ and width 2 cm . What is the perimeter of the rectangle?

$$\text{Length} = \frac{x}{2}$$

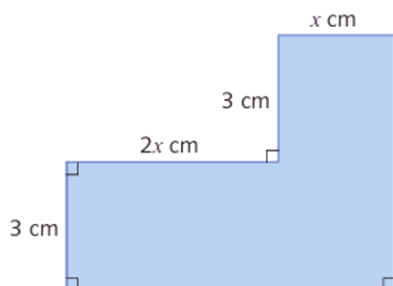
$$\text{Perimeter} = 2\left(\frac{x}{2} + 2\right) = (x + 4) \text{ cm}$$

- b A rectangle has area $(x + 3) \text{ cm}^2$ and width 2 cm . What is the perimeter of the rectangle?

$$\text{Length} = \frac{x + 3}{2}$$

$$\text{Perimeter} = 2\left(\frac{x + 3}{2} + 2\right) = (x + 7) \text{ cm}$$

- 7 For the shaded L -shaped region below, find:



- a the perimeter of the shaded region.

$$\text{Perimeter} = x + 3 + 2x + 3 + 3x + 6 = (6x + 12) \text{ cm}$$

- b the area of the shaded region.

$$\text{Width} = x + 6, \text{ height} = x + 4.$$

$$\text{Area} = (x + 6)(x + 4) \text{ cm}^2$$

- 8** Think of a number. Let this number be x .
- a** Multiply the number you thought of by 2 and subtract 5.

$$2x - 5$$

- b** Multiply the result by 3.

$$3(2x - 5) = 6x - 15$$

- c** Add 15.

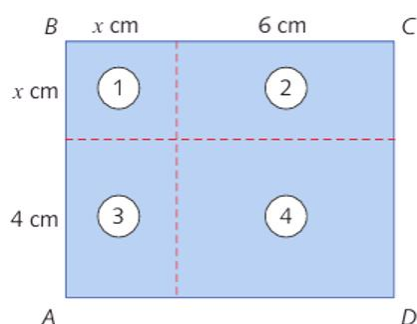
$$6x - 15 + 15 = 6x$$

- d** Subtract 5 times the number you first thought of. What is the final answer? Why is it so?

$$6x - 5x = x.$$

The result always equals the original number.

- 9** Rectangle $ABCD$ is divided into four smaller rectangles by a vertical and horizontal line.



- a** Find the area of each of the rectangles 1, 2, 3 and 4.

$$\text{Rectangle 1: } x \times x = x^2 \text{ cm}^2$$

$$\text{Rectangle 2: } 6 \times x = 6x \text{ cm}^2$$

$$\text{Rectangle 3: } x \times 4 = 4x \text{ cm}^2$$

$$\text{Rectangle 4: } 6 \times 4 = 24 \text{ cm}^2$$

- b** Find the area of rectangle $ABCD$ as a product of the lengths of its sides.

$$\text{Width} = x + 6, \text{ height} = x + 4. \text{ Area} = (x + 6)(x + 4) \text{ cm}^2$$

- c** Deduce that $(x + 6)(x + 4) = x^2 + 10x + 24$ for all values of x .

$$\text{Sum of the four areas: } x^2 + 6x + 4x + 24 = x^2 + 10x + 24.$$

$$\text{This also equals the area of } ABCD = (x + 6)(x + 4).$$

$$\text{Therefore } (x + 6)(x + 4) = x^2 + 10x + 24.$$

- d** Find the perimeter of each of the rectangles 1, 2, 3 and 4.

$$\text{Rectangle 1: } 4x \text{ cm}$$

$$\text{Rectangle 2: } 2(6 + x) = (12 + 2x) \text{ cm}$$

$$\text{Rectangle 3: } 2(x + 4) = (8 + 2x) \text{ cm}$$

$$\text{Rectangle 4: } 2(6 + 4) = 20 \text{ cm}$$

- e** Find the perimeter of rectangle $ABCD$.

$$2((x + 6) + (x + 4)) = (4x + 20) \text{ cm}$$

10 The area of rectangle $ABCD$ is 16 cm^2 . The length of BA is $x \text{ cm}$.



a Find the length of BC , in terms of x .

$$BC = \frac{16}{x}$$

b Write down an expression for the perimeter of the rectangle, in terms of x .

$$2\left(x + \frac{16}{x}\right) = \left(2x + \frac{32}{x}\right)$$

c If $x = 4$, find the perimeter of the rectangle.

$$2(4) + \frac{32}{4} = 16 \text{ cm}$$

d If $x = \frac{1}{4}$, find the perimeter of the rectangle.

$$2\left(\frac{1}{4}\right) + \frac{32}{\frac{1}{4}} = \frac{1}{2} + 128 = 128\frac{1}{2} \text{ cm}$$

11

a Fill in this table.

n	$n + 2$	$n - 2$	$(n + 2)(n - 2)$	n^2
2	4	0	0	4
3				
4				
5				
6				
7				

b Deduce that $(n + 2)(n - 2) = n^2 - 4$ for $n = 2, 3, 4, 5, 6$ and 7 .

In each row, $(n + 2)(n - 2) = n^2 - 4$. This holds for all values listed.

c Why does $n^2 - 4 = (n + 2)(n - 2)$ for all whole numbers n ?

$$\begin{aligned} & (n + 2)(n - 2) \\ &= n(n - 2) + 2(n - 2) \\ &= n^2 - 2n + 2n - 4 \\ &= n^2 - 4. \end{aligned}$$

Therefore $n^2 - 4 = (n + 2)(n - 2)$ for all whole numbers n .

12 Write an algebraic expression for each of the following:

a Find the area of the square whose perimeter is $12d$ cm.

$$\text{Side} = \frac{12d}{4} = 3d$$

$$\text{Area} = (3d)^2 = 9d^2 \text{ cm}^2$$

b Find the area of a rectangle whose perimeter is $32k$ cm and whose length is three times its width.

$$\text{Let width} = w, \text{ so length} = 3w$$

$$2(w + 3w) = 32k$$

$$w = 4k$$

$$\text{Area} = 3(4k)(4k) = 48k^2 \text{ cm}^2$$

c Find the equal length of an isosceles triangle whose perimeter is k cm and base d cm.

$$2l + d = k$$

$$l = \frac{k - d}{2} \text{ cm}$$

d Find the width of a rectangle whose perimeter is k cm and length d cm.

$$2(w + d) = k$$

$$w = \frac{k - 2d}{2} \text{ cm}$$

13 If $x = 2y$ and $y = 3z$ express:

a $x + 3y - 7z$ in terms of z

$$\text{Sub } x = 6z, y = 3z:$$

$$6z + 9z - 7z = 8z$$

b $x + 3y - 6z$

i in terms of y

$$\text{Sub } x = 2y, z = \frac{y}{3}:$$

$$2y + 3y - 6\left(\frac{y}{3}\right) = 3y$$

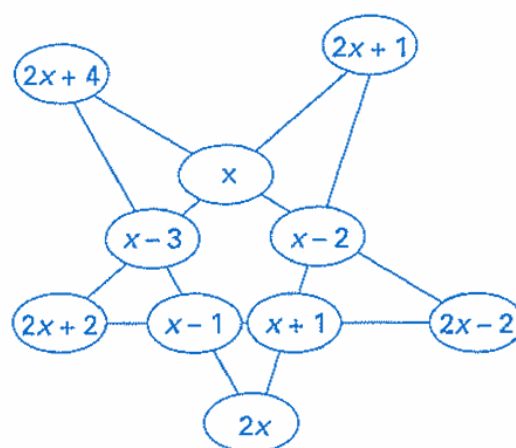
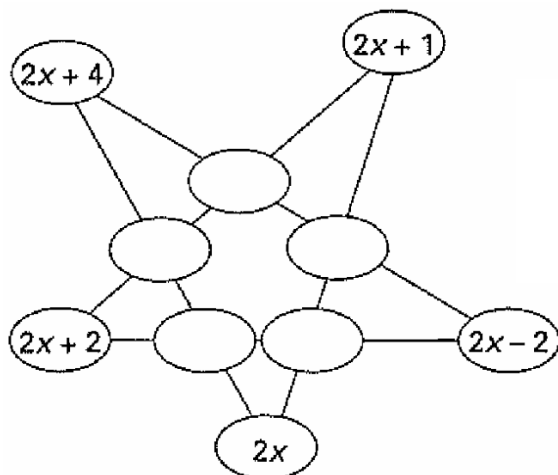
ii in terms of x

$$\text{Sub } y = \frac{x}{2}, z = \frac{x}{6}:$$

$$x + 3\left(\frac{x}{2}\right) - 6\left(\frac{x}{6}\right) = \frac{3x}{2}$$

14 Place the following algebraic expressions in the ovals so that four ovals in a row total $6x$.

$$x - 3, \quad x - 2, \quad x - 1, \quad x, \quad x + 1$$



- 15** A car travels 20 km from Cranung to Doville.
It travels d km at 60 km/h and the remainder at 80 km/h.

a How far does the car travel at 80 km/h?

$$(20 - d) \text{ km}$$

b For how long, in terms of d , does the car travel at:

i 60 km/h?

$$\frac{d}{60} \text{ h}$$

ii 80 km/h?

$$\frac{20 - d}{80} \text{ h}$$

c What is the total time for the journey, in terms of d ?

$$\begin{aligned} & \frac{d}{60} + \frac{20 - d}{80} \\ &= \frac{4d + 3(20 - d)}{240} \\ &= \frac{d + 60}{240} \text{ h} \end{aligned}$$

d Find the total time taken, in minutes, if:

i $d = 6$ km

ii $d = 12$ km

iii $d = 20$ km

$$\text{Convert to minutes: } \frac{d + 60}{240} \times 60 = \frac{d + 60}{4} \text{ min}$$

$$\text{i. } \frac{6 + 60}{4} = 16\frac{1}{2} \text{ min}$$

$$\text{ii. } \frac{12 + 60}{4} = 18 \text{ min}$$

$$\text{iii. } \frac{20 + 60}{4} = 20 \text{ min}$$

16

a If a is a whole number, what are the possible last digits of a^2 ?
(For example, if $a = 87$, the last digit is 9.)

Possible last digits of a^2 : 0, 1, 4, 5, 6, 9.

b If a is an even number, what are the possible values of the last digit of $a^2 + 1$?
(For example, if $a = 86$, the last digit of $a^2 + 1$ is 7.)

For even a , last digit of a is 0, 2, 4, 6, or 8.

Last digits of a^2 : 0, 4, 6, 6, 4.

Adding 1: 1, 5, 7, 7, 5.

Possible last digits of $a^2 + 1$: 1, 5, 7.

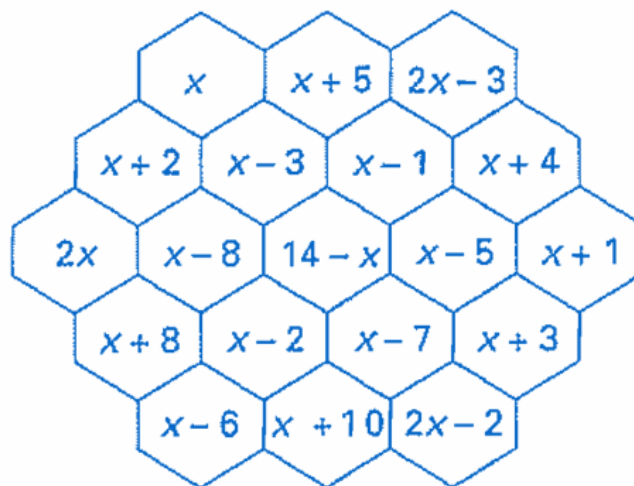
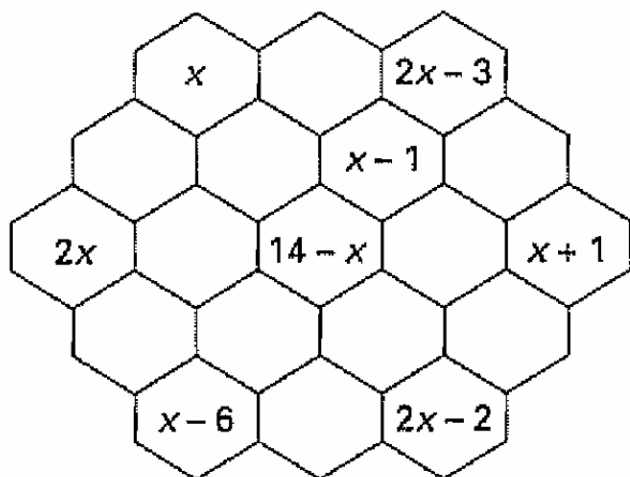
c If b is an even whole number divisible by 3, what are the possible last digits of b ?

b is even and divisible by 3, so b is divisible by 6.

Last digits of multiples of 6 cycle through: 6, 2, 8, 4, 0.

Possible last digits: 0, 2, 4, 6, 8.

- 17 Complete the hex grid so that any straight-line sums to $4x + 2$.



- 18 Expand each of the following:

a $2(3(4(5x + 2) + 3) + 4) + 5$

$$120x + 79$$

b $-2(-3(-4(-5x - 2) - 3) - 4) - 5$

$$120x + 33$$

c $4(3(2(x + 2) + 2) + 3) + 4$

$$24x + 88$$

d $-4(-3(-2(x - 2) - 3) - 4)$

$$-24x + 28$$

- 19 Write an expression for the following, then expand to show it always equals the same thing.

- Think of a number.
- Add 3 to your number.
- Multiply your answer by 2.
- Subtract twice your original number.
- Multiply your answer by 7.

For any initial value x , we get the expression $7(2(x + 3) - 2x)$.

$$= 7(2x + 6 - 2x)$$

$$= 7(6)$$

$$= 42$$